

Sardar Vallabhbhai National Institute of Technology Surat
Electronics Engineering Department
Subject: Digital Communication [EC303]
B.Tech III (Sem 5) Mid Sem Exam- October 2022

Time: 1.5 Hrs

Marks: 30

Attempt any 5 Questions: Each question carries 6 marks.

Q-1 A 7 bit PCM system employing uniform quantization has an overall signaling rate of 56 kbps. Calculate the signal to quantization noise that would result when its input is a sine wave with peak amplitude equal to 5 volt. For the same resolution, find the dynamic range for the sine wave inputs in order that the signal to quantization noise ratio may be less than 30 dB. What is the theoretical maximum frequency f_m that this system can handle? Based on channel SNR 35 dB what is the channel capacity? [Hint: signaling rate $r = \text{number of bits } n \times \text{sampling rate } f_s$] [6]

Q-2

- (a) What do you mean by aperture effect in flat top sampling? How to eliminate it? [2]
- (b) Draw the model of Binary Symmetric Channel. [1]
- (c) An instantaneous adaptive delta modulator adapts its step size exponentially. i.e. if the predictor is unable to track the input, the step size is doubled. Thus the steps generated have the values given by $S_n = 2^n$, where $n = 1, 2, 3, \dots, 10$. Find the Entropy of the step generator if the probability of a step S_n is $p(S_n) = 1/S_n$ for $n = 1, 2, 3, \dots, 9$ and $p(S_{10}) = p(S_9)$. [3]

Q-3 Digital voice of 32 kbps is to be transmitted through a bandwidth of 3.1 kHz. The transmitted power is such that the received signal to noise ratio is 1200. Determine if the data can be transmitted error free through this channel. If not, then determine the minimum SNR required for error free transmission and spectrum efficiency in this case. What is E_b/N_0 for the received data in this case? If E_b/N_0 is to achieve the Shannon limit, what must be S/N ? [6]

Q-4 A DMS X has five symbols x_1, x_2, x_3, x_4, x_5 with $p(x_1) = 0.4, p(x_2) = 0.19, p(x_3) = 0.16, p(x_4) = 0.15$ and $p(x_5) = 0.1$.

- (i) Construct a Shannon-Fano code for X and calculate the efficiency of the code.
- (ii) Repeat for the Huffman coding and compare the results. [6]

Q-5 (a) Mathematically prove that $H(X, Y) = H(X) + H(Y/X)$ as per information theory with proper assumptions. [3]

(b) Explain in detail the synchronous and asynchronous transmissions with the frame formats generated in each case. [3]

Q-6 An analog signal is bandlimited to 6 kHz and sampled at Nyquist rate. The samples are quantized into 4 levels. Each level represent one symbol. The probabilities of occurrence of these 4 levels are $p(x_1) = p(x_4) = 1/8, p(x_2) = p(x_3) = 3/8$. Obtain information rate R of this source. If this source would be an equiprobable DMS what would be the information rate of the source? If the AWGN channel SNR is 42 dB find the theoretical limit of the bit rate C so that $H(X) \leq R \leq C$. [6]